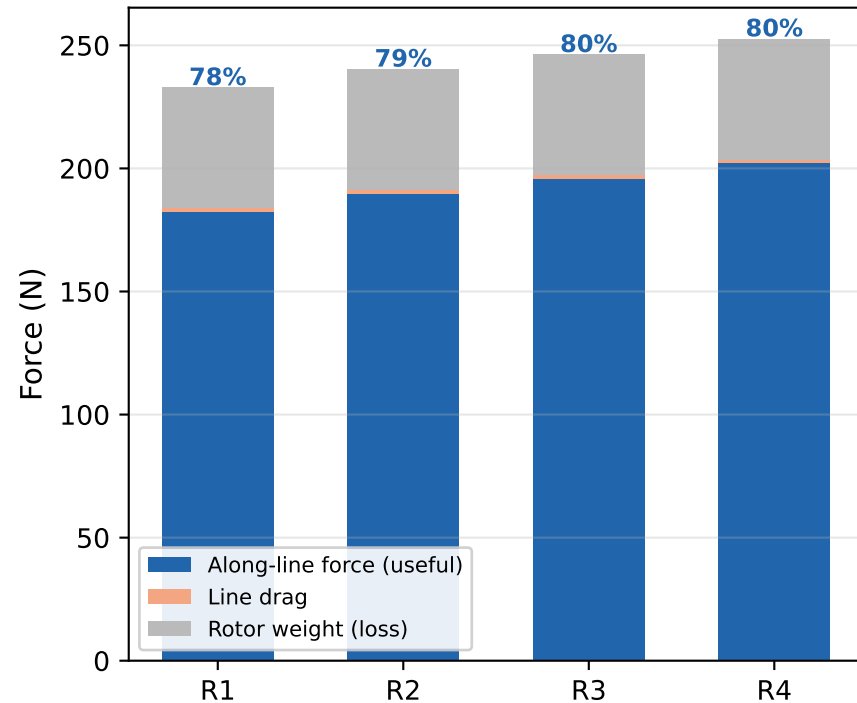
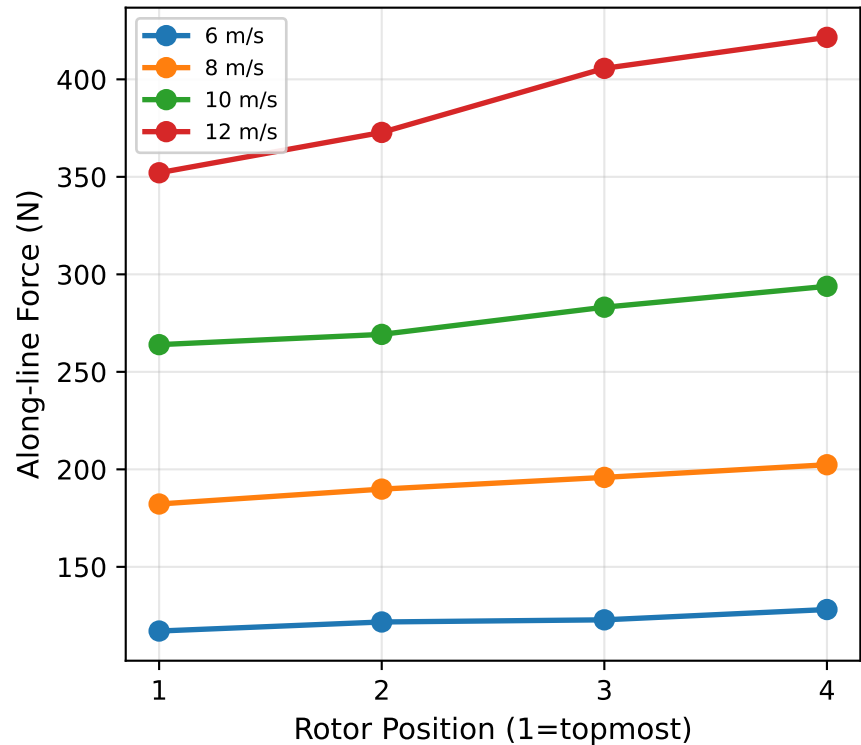


Force Budget per Rotor
R=3.0m, N=4, 8 m/s



Along-line Force vs Position
R=3.0m, N=4, Graded Tilt, 45°



IMPLICATIONS: Rotor weight (49 N) is the dominant loss — 20-27% of thrust. Line drag is negligible (<1 N per section). Along-line efficiency improves with wind speed (73%→79%). Force increases from top to bottom rotor by 10% — the polygon chain gives lower rotors a more favorable inflow angle. Data: BEM v2.1, corrected polygon solver. See SPEC.md §6.6.